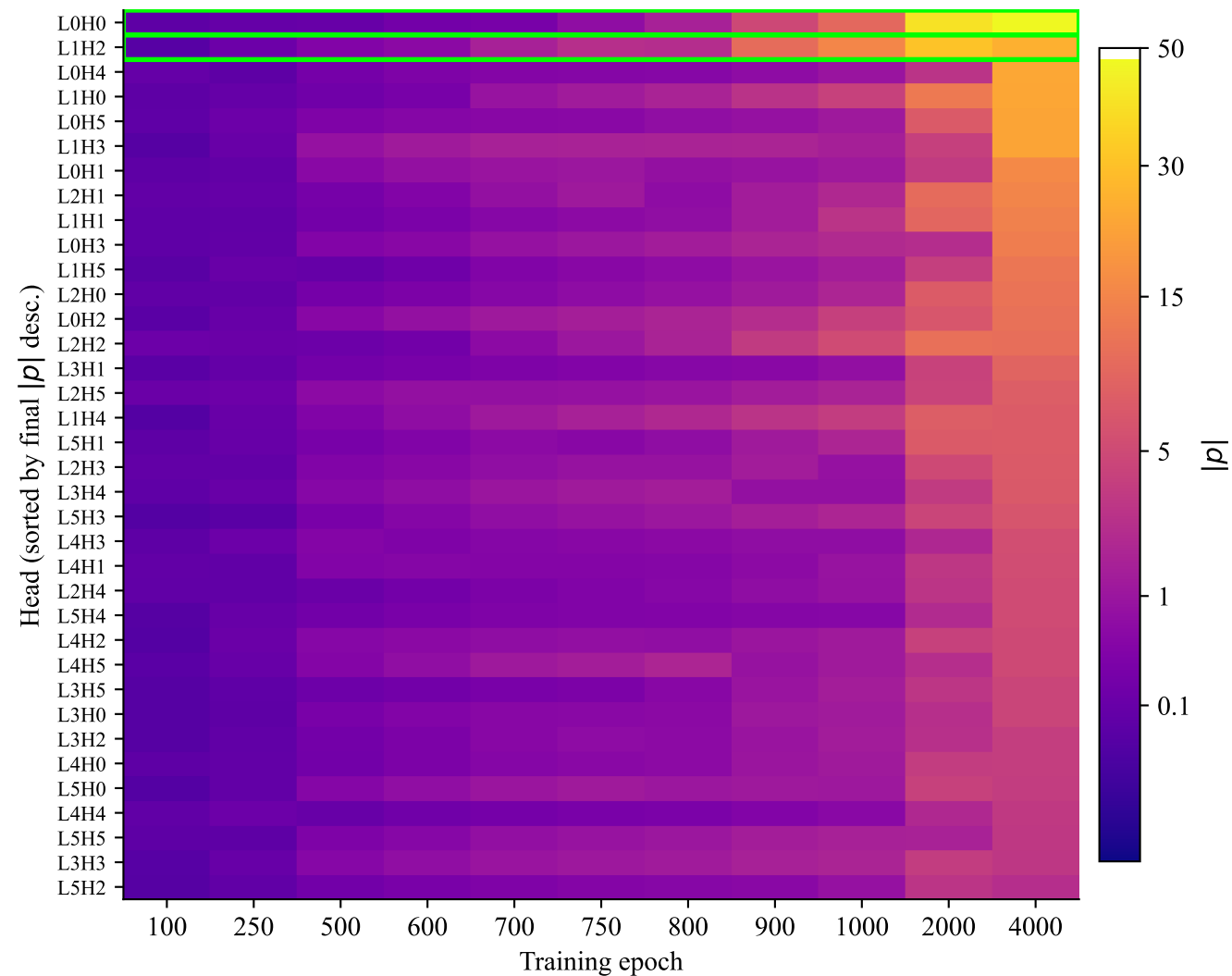


Spatial-relation alignment across training: L_0H_0 and L_1H_2 emerge sharply

(a) Per-head emergence: projection magnitude $|p|$



(b) Phase-transition emergence at epoch 600–800

